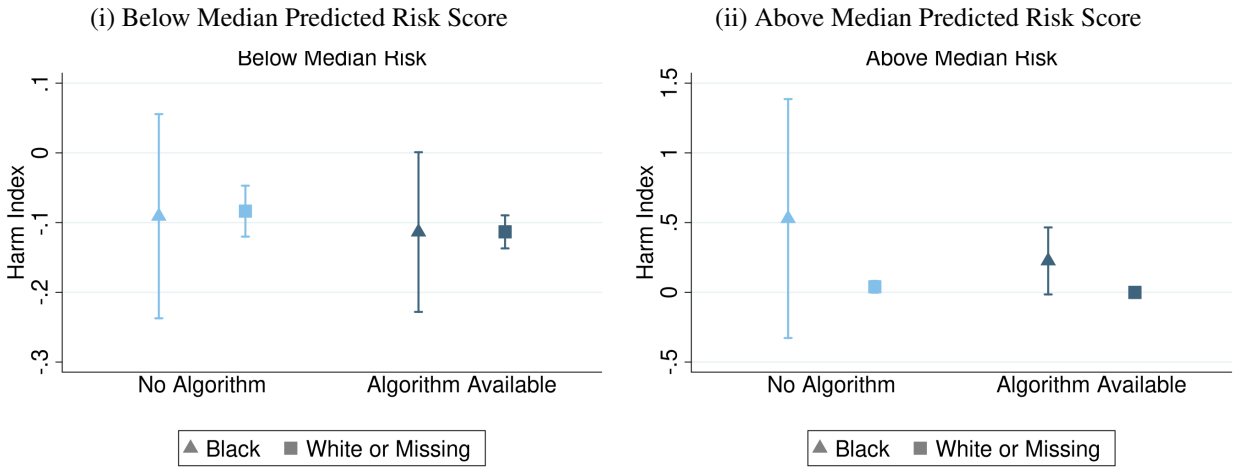
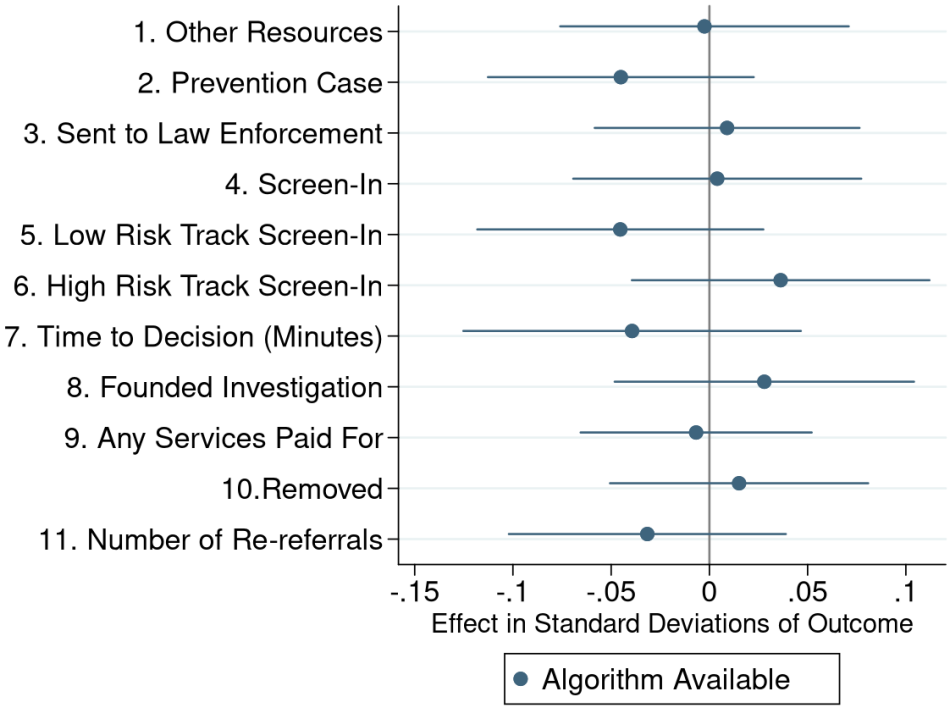
Figure A.10: No Significant Increase in Harm from Reducing Screen-In Disparities

Using CPS-Known Race



Notes: These figures show the estimates of mean harm index by demographics, tool availability, and whether algorithm-predicted risk was above or below median. Figures define race as recorded by Child Protective Services. 90% confidence intervals are reported and randomization controls are included throughout.

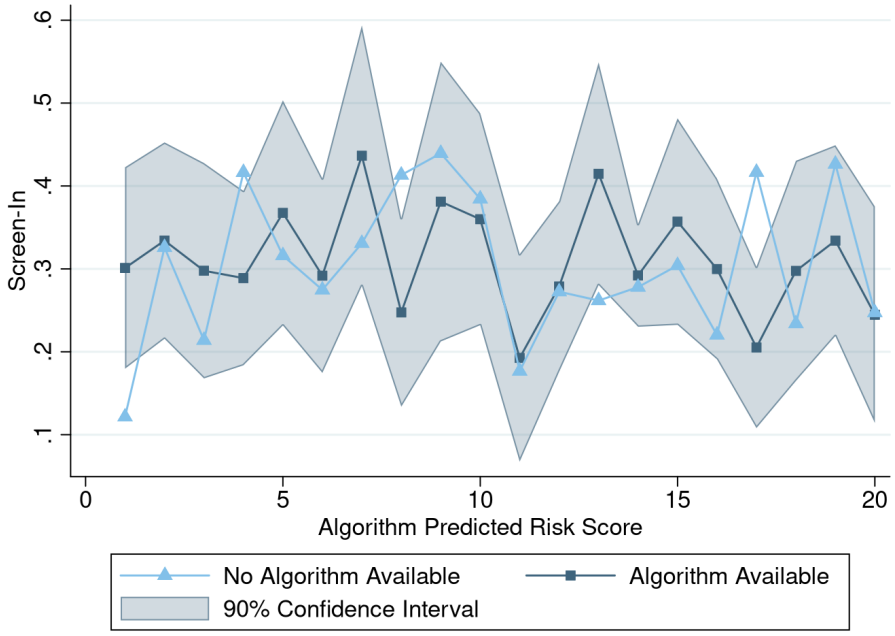
Figure A.11: Level Effect of Tool on Margins of CPS Intervention



Notes: This figure presents point estimates and confidence intervals of regressing the listed outcomes on tool availability, as in equation (1), for the full sample of children (unless otherwise specified). Estimates are presented with standardized outcome variables (mean 0, variance 1) for comparability of coefficient magnitudes across different outcomes. We measure margins of CPS intervention, such as high-risk-track screen-ins, using child welfare administrative data. Additional margins of intended intervention, such as voluntary resources sent to screened-out families, are inferred from team discussion notes. Some interventions are unobserved: services paid for by Medicaid, for example, are not included in administrative data. Workers can recommend opening a prevention case (row 2), can send the referral to law enforcement (row 3), and can connect the family to other resources

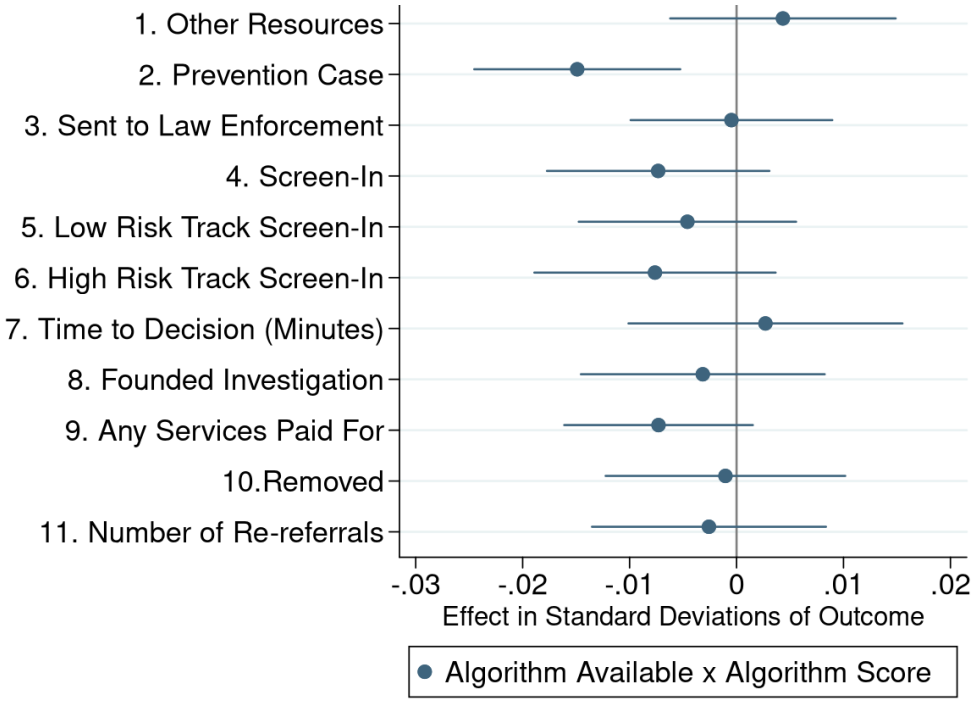
(row 1) instead or in addition to screening in (row 4). If workers decide to screen in a family, they also choose whether the screen-in is on a low-risk or high-risk track (rows 5 and 6). Low-risk track investigations are a type of differential response that is less intrusive for the family. Formal investigations are only run for the high-risk track. As a result, having a founded (i.e., substantiated; row 8) investigation requires having a high-risk track investigation. Row 9 shows the change in any other services paid for directly by CPS. The number of minutes it took workers to make a decision (row 7) is measured using time stamp data and is only available for observations where edits to the notes were not made after normal discussion meeting times. The final two rows (rows 10-11) report estimates on whether a child was ever removed to foster care, and the number of re-referrals. We report more details on these last two outcomes in a separate table (Appendix Table A.7). Randomization procedure controls are included throughout. Standard errors are clustered at the household level and are shown at the 90 percent confidence level.

Figure A.12: Workers Did Not Increase Investigations for High-Scoring Children



Notes: This figure presents screen-in rates with and without the algorithm tool, by algorithm predicted risk score. It includes the full sample of child referrals. For visibility purposes, 90% confidence intervals are displayed for the intervention group only. Randomization controls are included throughout.

Figure A.13: Effect of Tool on Margins of CPS Intervention, Interacted with Algorithm Score



Notes: This figure presents point estimates and confidence intervals of regressing the listed outcomes on tool availability interacted with algorithm score for the full sample of children (unless otherwise specified). We report point estimates and standard errors for the coefficient on the interaction of tool availability with algorithm score. Following our pre-registration plan, we use the maximum score of all children on the referral, as this maximum score was most salient to workers and most of the decisions listed typically apply to all children in a household. Estimates are presented with standardized outcome variables (mean 0, variance 1) for comparability of coefficient magnitudes across different outcomes. We measure margins of CPS intervention, such as high-risk-track screen-ins, using child welfare administrative data. Additional margins of intended intervention, such as voluntary resources sent to screened-out families, are inferred from team discussion notes. Some interventions are unobserved: services paid for by Medicaid, for example, are not included in administrative data. Workers can recommend opening a prevention case (row 2), can send the referral to law enforcement (row 3), and can connect the family to other resources (row 1) instead or in addition to screening in (row 4). If workers decide to screen in a family, they also choose whether the screen-in is on a low-risk or high-risk track (rows 5 and 6). Low-risk track investigations are a type of differential response that is less intrusive for the family. Formal investigations are only run for the high-risk track. As a result, having a founded (i.e., substantiated; row 8) investigation requires having a high-risk track investigation. Row 9 shows the change in any other services paid for directly by CPS. The number of minutes it took workers to make a decision (row 7) is measured using time stamp data and is only available for observations where edits to the notes were not made after normal discussion meeting times. The final two rows (rows 10-11) report estimates on whether a child was ever removed to foster care, and the number of re-referrals. We report more details on these last two outcomes in a separate table (Appendix Table A.7). Randomization procedure controls are included throughout. Standard errors are clustered at the household level and are shown at the 90 percent confidence level.